
Adaptive Visual Autoregressive Acceleration via Dual-Linkage Entropy Analysis

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Figure 1. NOVA generates a 2K image in 3.2 seconds on a single NVIDIA RTX 3090 GPU without running out of GPU memory.

Abstract

Visual AutoRegressive modeling (VAR) suffers from substantial computational cost due to the massive token count involved. Failing to account for the continuous variation in modeling, existing VAR token reduction methods face three key limitations: heuristic stage partition, non-adaptive schedules, and limited acceleration scope, thereby leaving significant acceleration potential untapped. Since entropy variation intrinsically reflects the transition of predictive uncertainty, it offers a principled measure to capture continuous modeling variation. Therefore, we propose NOVA, a training-free token reduction acceleration framework for VAR models via entropy analysis. NOVA adaptively determines the acceleration activation scale during inference by online identifying the inflection point of scale entropy growth. Through scale-linkage and layer-linkage ratio adjustment, NOVA dynamically computes distinct token reduction ratios for each scale and layer, pruning low-entropy tokens while reusing

the cache derived from the residuals at the prior scale to accelerate inference and maintain generation quality. Extensive experiments and analyses validate NOVA as a simple yet effective training-free acceleration framework. Code is [available](#).

1. Introduction

Recently, Visual AutoRegressive modeling (VAR) (Tian et al., 2024) based on next-scale prediction has demonstrated impressive breakthroughs in visual generation, achieving performance comparable to mainstream diffusion models (Shen et al., 2025; Croitoru et al., 2023; Cao et al., 2024). However, VAR’s performance improvement comes at the expense of a substantial computational cost. As illustrated in Figure 2, its inefficiency mainly stems from the considerable computation complexity caused by the excessive token count, which further hinders its scalability to higher-resolution visual generation and integration into other downstream tasks.

Towards direct and efficient application of VAR, training-free token reduction methods (Guo et al., 2025a; Chen et al., 2025b; Li et al., 2025d) have emerged as promising acceleration solutions. These methods build on a common observation: VAR’s substantial computational cost mainly occurs in the later stage. Consequently, they perform token reduction at this stage to accelerate inference. Although these methods achieve impressive results, they still face

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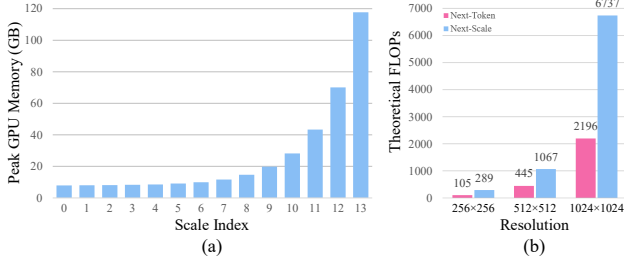


Figure 2. (a) Peak GPU memory consumption across different scales during 1024×1024 image generation in VAR. As the scale increases, the token sequence length grows rapidly, leading to quadratic growth in attention computation complexity. (b) Theoretical GFLOPs required by next-token and next-scale prediction models to generate images of various resolutions under uniform settings. Compared to the raster-scan next-token prediction, VAR’s coarse-to-fine next-scale prediction additionally requires predicting feature maps at multiple previous scales before generating the final scale. The cumulative computation at these previous transition scales introduces extra computational cost to VAR.

three key limitations: (1) *Heuristic stage partition*. The later stage partition heavily relies on statistical estimation or empirical observation. The precision of this boundary is critical: premature token reduction risks quality degradation, while delayed token reduction limits acceleration gains. (2) *Non-adaptive schedules*. All instances share the fixed stage boundary and token reduction ratio. However, different instance images vary enormously in complexity regarding semantics, composition, structure, lighting, and texture. The fixed stage partition and reduction ratio cannot guarantee optimality for every instance. (3) *Limited acceleration scope*. Token reduction is performed only in the partitioned later stage, leaving potential computational redundancy in earlier stages unaddressed.

We attribute above limitations to a fundamental oversight: treating scales and layers as isolated units while neglecting the continuous variation in model predictions across scales and layers. Specifically, these methods rigidly partition the later stage and independently optimize the token reduction objectives for each scale or layer in the later stage. However, VAR inherently follows a coarse-to-fine transitional process that necessitates consideration of continuous modeling evolution across both scales and layers. Thus, we argue that it is essential to adopt a dual-linkage (scales and layers) adaptive token reduction method.

From the viewpoint of information theory, autoregressive generation is essentially a process of progressively reducing sequential uncertainty through successive observations. Each token generation represents a new observation. As a direct measure of uncertainty, entropy effectively reflects the change in predictive uncertainty throughout the process. A high-entropy token signifies a highly uncertain prediction with the newly observed information that may significantly impact subsequent distribution predictions, typically corresponding to larger potential for entropy reduction. There-

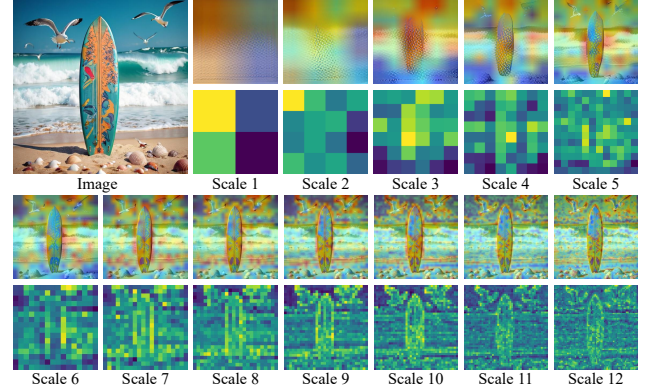


Figure 3. Entropy heatmaps of visualized images and token maps.

fore, accurately modeling it offers substantial information gain¹ to further reduce sequential predictive uncertainty. Conversely, low-entropy tokens correspond to confident predictions for the current states with limited marginal impact on further narrowing distributions. Moreover, entropy variation allows us to analyze the uncertainty reduction process across scales or layers, thereby capturing the evolution of modeling dynamics. Meanwhile, as visualized in Figure 3, entropy heatmaps intuitively reveal the spatio-temporal diffusion characteristics of high-entropy regions: evolving from a globally concentrated pattern in early scales to a sparsely dispersed pattern in later scales. These analyses indicate that entropy can naturally serve as a principled token selection criterion for our dual-linkage adaptive token reduction method.

In light of the above discussion, we propose NOVA, a training-free acceleration framework for VAR models. Guided by entropy analysis, NOVA performs a dual-linkage adaptive token reduction across both scales and layers. Specifically, at the scale level, it adaptively determines the scale for acceleration activation by online identifying the inflection point in scale entropy growth curve, and computes a distinct token reduction ratio for each acceleration scale by a scale-linkage function. At the layer level, NOVA further utilizes a layer-linkage ratio adjustment function to determine the specific reduction ratio for each layer and reuses the cache from residual interpolation of the same layer at the prior scale to improve speed and maintain performance.

Extensive experiments and analysis demonstrate the effectiveness of NOVA. On GenEval, NOVA achieved a 2.89× speedup for Infinity-2B with only 0.01% performance loss, while surpassing comparable methods in both speed and performance. On ImageReward, NOVA not only reduced Infinity-8B’s latency from 1.51s to 0.75s, but even achieved a preference score surpassing the original Infinity-8B. We further provide extensive visualizations to demonstrate the

¹Information gain, a fundamental concept in information theory, quantifies the reduction in predictive uncertainty.

NOVA’s excellent preservation of details and semantics.

2. Related Work

Next-Scale Visual AutoRegressive Generation. By converting raster-scan order prediction to next-scale prediction, Visual AutoRegressive modeling (VAR) (Tian et al., 2024) overcomes the limitations of next-pixel prediction and next-token prediction, which suffer from suboptimal performance due to disrupting the inherent spatial structure of images. Next-scale prediction and elevates generation quality to new heights. Some VAR-like variants, such as Infinity (Han et al., 2025), HART (Tang et al., 2024), STAR (Ma et al., 2024) and VAR-CLIP (Zhang et al., 2024) extend VAR to text-to-image generation. Other improved VAR-like models (Li et al., 2024b; Rajagopalan et al., 2025; Xie et al., 2024; Wu et al., 2025; Jiao et al., 2025; Qu et al., 2025; Wang et al., 2025) effectively perform various tasks.

Efficient Visual AutoRegressive Modeling. To address VAR’s efficiency bottleneck, numerous efforts (Huang et al., 2025; He et al., 2025; Kumar et al., 2025; Zhang et al., 2025b;a) have been made. However, all these methods necessitate major architectural modifications (Ren et al., 2024; Chen et al., 2025c; Vincenti et al., 2025) or extensive retraining (Tang et al., 2024; Kumbong et al., 2025), making them difficult to directly obtain and deploy. Training-free methods, such as FreqExit/SkipVAR’s early-exit inference or decision-based acceleration (Li et al., 2025b; 2026), ScaleKV/HACK’s KV-Cache optimization (Li et al., 2025c; Qin et al., 2025), and LiteVAR’s efficient attention and low-bit quantization (Xie et al., 2024), achieve good speedups but face some challenges, such as generation quality degradation due to skipping entire layers or blocks instead of individual tokens and suboptimal performance due to input-independent static acceleration. In contrast, training-free token reduction methods (Guo et al., 2025a; Chen et al., 2025b; Li et al., 2025d; Kou et al., 2026a) have led the development of efficient VAR due to their strong generalizability and flexibility. However, due to the lack of adaptivity and dynamics, their acceleration potential remains unleashed.

Entropy-Guided Deep Learning. Entropy, as a measure of uncertainty and information distribution, is inherently linked to probabilistic modeling. Consequently, entropy-guided methods have been widely adopted in deep learning. Studies (Peer et al., 2022; Meni et al., 2024; Dubey et al., 2017) utilize entropy-based regularization to improve robustness and stability. SEGA (Wu et al., 2023a), MGEDE (Yang et al., 2023) and SEBot (Yang et al., 2024) enhance graph representation learning from an entropy perspective. EGGesture (Xiao et al., 2024) and EDRL (Wang et al., 2023) propose entropy-guided methods for VQ-VAE and disentangled representation learning, respectively. Similar entropy prin-

ciples also benefit label distribution learning ((Kou et al., 2024; 2025a;c;b; 2026b; Wu et al., 2026)). Meanwhile, entropy is also employed to improve efficiency, such as accelerating robotic visuomotor policies (Guo et al., 2025b), video diffusion models (Li et al., 2025a), and diffusion language models (Ben-Hamu et al., 2025). However, its application to visual autoregressive generation almost leaves blanks.

3. Methodology

3.1. Preliminary

VAR (Tian et al., 2024) reformulates autoregressive modeling from next-token prediction to next-scale residual feature prediction, generating images in a coarse-to-fine manner. Formally, the modeling process involves T multi-scale token maps $\{R_1, R_2, \dots, R_T\} = \{R_t\}_{t=1}^T$, where $R_t \in [V]^{h_t \times w_t}$ and (h_t, w_t) denotes the grid size at scale t . At the t -th scale, VAR predicts the token map R_t conditioned on all previous scales $R_{<t} = \{R_1, R_2, \dots, R_{t-1}\}$. The autoregressive likelihood is formulated as follow:

$$p(R_1, R_2, \dots, R_T) = \prod_{t=1}^T p(R_t \mid \langle \text{sos} \rangle, R_{<t}), \quad (1)$$

where $\langle \text{sos} \rangle$ denotes the start conditional embedding.

3.2. Analysis

Theoretical Foundation for Entropy-Guided Token Reduction. In information theory, given a conditioning variable Z , the conditional mutual information between random variables X and Y satisfies

$$I(X; Y \mid Z) \leq H(X \mid Z), \quad (2)$$

where $H(X \mid Z)$ denotes the conditional entropy of X given Z , and conditional mutual information $I(X; Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z)$ measures information gain, i.e., the reduction in uncertainty of Y after observing X given Z .

VAR treats scales as autoregressive steps. For t -th scale, let $N_t = h_t w_t$ and denote the token at R_t spatial index $i \in \{1, \dots, N_t\}$ as $r_{t,i} \in [V]$. Therefore, the upper bound of subsequent uncertainty at the t -th scale is determined by the sum of conditional entropies of all tokens in R_t :

$$H(R_t \mid R_{<t}) \leq \sum_{i=1}^{N_t} H(r_{t,i} \mid R_{<t}). \quad (3)$$

Combining Equations 2 and 3, it follows that maintaining highest-entropy tokens ensures the upper bound of uncertainty remains controllable while maximizing potential information gain to reduce future uncertainty, thereby facilitating the autoregressive generation process. Meanwhile,

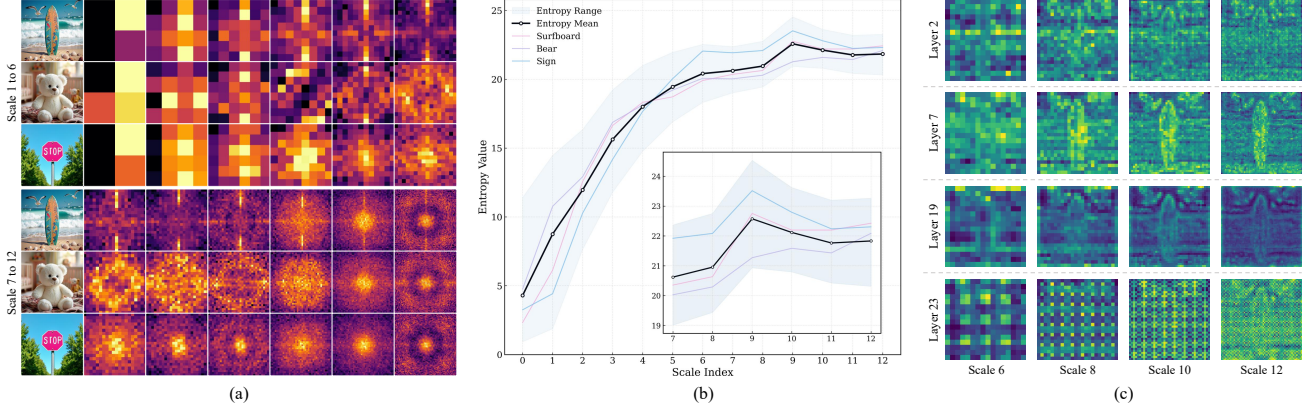


Figure 4. (a) Frequency spectrum of various instances at different scales during VAR generation. (b) Various entropy values across different scales during the inference process of Infinity-2B: We selected 600 prompts from GenEval and rewrote them. For each rewritten prompt, we generated four images using different random seeds. (c) Entropy heatmaps of token maps across different Transformer layers. Cross-layer spatial pattern variations: Shallow layers exhibit high-entropy regions resembling local noise or scattered dots, certain middle layers show high-entropy regions with clear subject contours, while deeper layers display regular periodic grid-like patterns in high-entropy regions. Cross-layer overall entropy variations: For example, token map at the 19-th layer has relatively low overall entropy.

pruning the lowest-entropy tokens has limited impact on potential information gain but effectively reduces attention computation cost, thus providing theoretical foundation for our entropy-guided token reduction method.

Scale-Level Generation Process Analysis. Existing methods (Guo et al., 2025a; Chen et al., 2025b; Li et al., 2025d) employ frequency to heuristically partition the later stage and perform token reduction within it. As shown in Figure 4 (a), we analyze the frequency spectrum evolution of token maps at different scales during generation across the three instances. Although a general coarse-to-fine process and certain stage-specific characteristics are observable, the distribution, content, and diffusion evolution of frequency components vary significantly across instances, making it difficult to adaptively and accurately partition the later stage through frequency analysis. Furthermore, some instances exhibit high-frequency patterns even in the middle scales, indicating more opportunities for token reduction prior to the later stage. While these works are highly novel and interesting and have achieved impressive results, we realize that relying solely on frequency, the physical property of images, makes it difficult to precisely capture the modeling process. Only by addressing the limitations of heuristic stage partition, non-adaptive schedules, and limited acceleration scope from the intrinsic properties of VAR generation, its acceleration potential can be further unleashed.

We calculate the predictive entropy for all instances across scales. Figure 4 (b) illustrates the overall entropy range and mean, while highlighting the entropy variations for the three aforementioned instances: surfboard, bear, and sign. The subfigure reveals that the entropy of three instances fluctuates and diverges from one another at the larger scales,

significantly deviating from the entropy mean. This suggests that even from an entropy perspective, it remains challenging to accurately and adaptively partition the later stage. Unlike previous works, this frees us from the focus on later stage partition. Nevertheless, the entropy range, the entropy mean, and the three instances all exhibit a consistent two-stage dynamic: entropy grows rapidly at early scales, followed by a phase of relatively slow growth or unstable fluctuation. This aligns with previous findings (Li et al., 2025d; Chen et al., 2025a;c) that VAR models construct concepts and objects at early scales. This motivates leveraging the entropy growth inflection point to adaptively activate acceleration, avoiding quality degradation from premature reduction and limited acceleration from delayed reduction.

Layer-Level Generation Process Analysis. We further analyze the entropy characteristics of token maps across different Transformer layers at the same scale. Figure 4 (c) reveals significant layer-level entropy heterogeneity. First, different layers exhibit various spatial distributions of high-entropy regions, indicating their distinct spatial modeling patterns. Second, the overall (average) entropy at the same scale also varies across layers. These variations suggest a uniform reduction ratio across all layers is suboptimal. Instead, token reduction at the same scale should be layer-adaptive, allocating more computation to high-entropy layers while reducing more tokens in low-entropy layers.

3.3. NOVA

Figure 5 illustrates the NOVA framework at both scale-level and layer-level, which consists of three key components: Adaptive Acceleration Activation, Dual-Linkage Acceleration and Residual Cache Reuse.

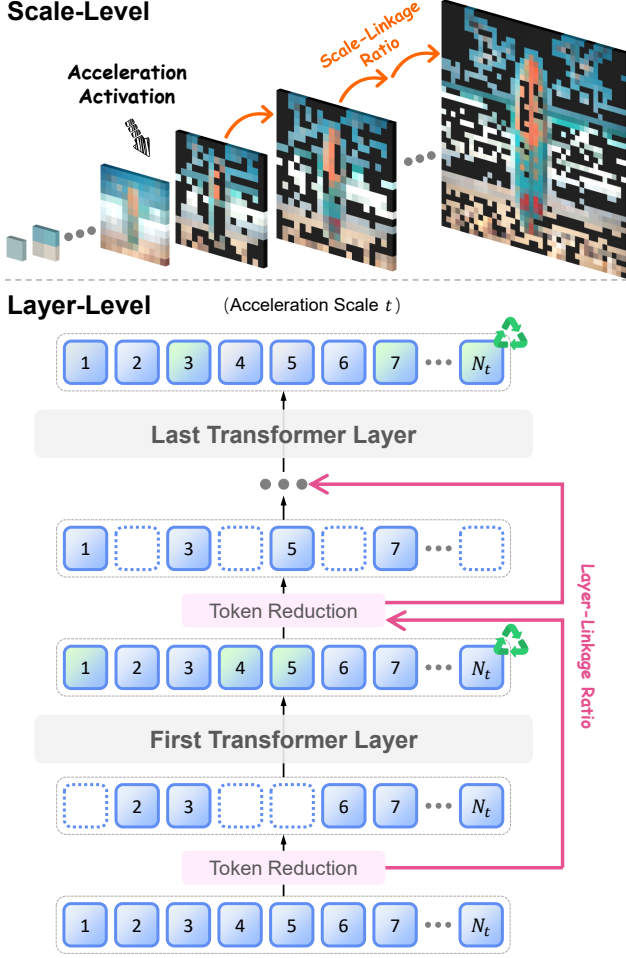


Figure 5. NOVA framework at both scale-level and layer-level.

Adaptive Acceleration Activation. At the scale-level, NOVA first determines the acceleration activation scale by online identifying the inflection point of the entropy growth across scales. For the token map R_t at the t -th scale, the predictive entropy of the i -th token $r_{t,i} \in [V]$ is defined as:

$$\mathcal{H}_{t,i} = H(r_{t,i} | R_{<t}) = - \sum_{v \in [V]} p_{t,i}(v) \log p_{t,i}(v), \quad (4)$$

where $p_{t,i}(\cdot)$ is the predicted class distribution for token $r_{t,i}$. Since VAR performs parallel prediction for N_t tokens within the t -th scale, we adopt a factorized conditional likelihood within the token map R_t :

$$p(R_t | R_{<t}) = \prod_{i=1}^{N_t} p(r_{t,i} | R_{<t}). \quad (5)$$

Therefore, the joint predictive entropy of the token map R_t decomposes additively into token predictive entropy:

$$H(R_t | R_{<t}) = \sum_{i=1}^{N_t} H(r_{t,i} | R_{<t}) = \sum_{i=1}^{N_t} \mathcal{H}_{t,i}. \quad (6)$$

We aggregate the predictive entropies of all tokens in the token map R_t and calculate their mean to obtain the t -th scale overall predictive entropy $\bar{\mathcal{H}}_t$:

$$\bar{\mathcal{H}}_t \triangleq \frac{1}{N_t} H(R_t | R_{<t}) = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathcal{H}_{t,i}. \quad (7)$$

In our earlier analysis, we found that the scale overall entropy $\bar{\mathcal{H}}_t$ exhibits a consistent two-stage dynamic as the scale increases. Therefore, using $\bar{\mathcal{H}}_t$ as a criterion, we online detect the decay of the entropy growth rate to identify the earliest scale where the growth rate decays and remains small as the acceleration activation scale. We define the discrete entropy growth rate g_t at scale t as:

$$g_t \triangleq \bar{\mathcal{H}}_t - \bar{\mathcal{H}}_{t-1}, \quad t = 2, \dots, T. \quad (8)$$

As a training-free acceleration framework, NOVA cannot access the overall entropy of all future scales at the beginning of inference. Therefore, we preserve the first t_{est} scales without token reduction to estimate the early entropy growth baseline for the current instance:

$$\eta \triangleq \frac{1}{t_{\text{est}} - 1} \sum_{t=2}^{t_{\text{est}}} g_t, \quad (9)$$

Considering the volatile fluctuations in $\bar{\mathcal{H}}_t$, we perform a moving average $\tilde{g}_t$ on two adjacent discrete entropy growth rates g_t obtained after the early entropy growth baseline estimation to mitigate misjudgments caused by oscillations: $\tilde{g}_t = (g_t + g_{t-1})/2$, where $t \geq t_{\text{est}} + 1$.

We identify the earliest scale t^* when the smoothed growth rate drops below a fraction $\alpha \in (0, 1)$ of the early entropy growth baseline η :

$$t^* \triangleq \min \{t \mid t \geq t_{\text{est}} + 1, \tilde{g}_t \leq \alpha \eta\}. \quad (10)$$

The scale t^* is regarded as the inflection point of the scale overall entropy growth. Consequently, NOVA activates acceleration at the scale $t^* + 1$.

Dual-Linkage Acceleration. The core of dual-linkage acceleration lies in dynamically adjusting and determining the distinct token reduction ratio for each scale or layer based on entropy variations of token maps. We design scale-linkage and layer-linkage ratio functions to realize.

For scale-linkage ratio function, we first assign each scale a base reduction ratio that progressively increases as generation proceeds. We then refine this base ratio by entropy growth rate g_t : if uncertainty is still changing rapidly, we reduce fewer tokens; if it trends toward stability, we reduce more. Thus, the token reduction ratio at the t -th scale is:

$$\text{Ratio}_t = \sigma \left(\frac{t - (t^* + 1)}{\tau} \right) - \lambda \cdot \tanh(g_t), \quad (11)$$

Table 1. Quantitative comparisons of perceptual quality on the GenEval and DPG-Bench benchmarks.

Methods	Speed $\uparrow$	Latency $\downarrow$	Param $\downarrow$	GenEval $\uparrow$				DPG-Bench $\uparrow$		
				Two Obj.	Position	Color Attr.	Overall	Global	Relation	Overall
HART	1.00 $\times$	1.12s	0.7B	0.62	0.13	0.18	0.51	87.10	92.95	80.89
+FastVAR	1.30 $\times$	0.86s	0.7B	0.57	0.16	0.19	0.50	86.21	92.48	80.71
+SparseVAR	1.38 $\times$	0.81s	0.7B	0.60	0.12	0.21	0.50	86.63	92.54	80.70
+NOVA	1.62 $\times$	0.69s	0.7B	0.60	0.13	0.17	0.51	86.88	92.53	80.84
Infinity-2B	1.00 $\times$	2.43s	2.0B	0.85	0.45	0.54	0.73	85.10	92.37	83.12
+FastVAR	2.45 $\times$	0.99s	2.0B	0.81	0.37	0.52	0.70	84.41	92.76	82.86
+SparseVAR	1.66 $\times$	1.46s	2.0B	0.84	0.42	0.54	0.72	84.11	92.70	82.56
+SkipVAR	2.23 $\times$	1.09s	2.0B	0.84	0.39	0.60	0.72	84.19	93.07	82.94
+NOVA	2.89 $\times$	0.84s	2.0B	0.86	0.42	0.58	0.72	84.36	92.68	82.93
Infinity-8B	1.00 $\times$	1.51s	8.0B	0.94	0.58	0.68	0.80	85.10	94.50	86.60
+FastVAR	1.26 $\times$	0.79s	8.0B	0.92	0.55	0.68	0.78	84.80	92.30	86.46
+SkipVAR	1.22 $\times$	0.82s	8.0B	0.91	0.57	0.69	0.77	83.91	90.52	86.40
+NOVA	1.33 $\times$	0.75s	8.0B	0.92	0.58	0.69	0.79	84.27	93.61	86.53

where $\sigma(\cdot)$ is the sigmoid function, λ is a scaling factor.

For layer-linkage ratio function, we dynamically refine the scale reduction ratio by computing the relative deviation between the current layer’s overall entropy and the mean overall entropy of all preceding layers and use this as the criterion for adjusting the token reduction ratio at the current layer. For the t -th scale and j -th layer, the mean overall entropy $\mu_{t,j-1}$ of the preceding $j-1$ layers is:

$$\mu_{t,j-1} = \frac{1}{j-1} \sum_{k=1}^{j-1} \bar{\mathcal{H}}_{t,k}, \quad j \geq 2. \quad (12)$$

Therefore, $\Delta_{t,j} = (\bar{\mathcal{H}}_{t,j} - \mu_{t,j-1}) / \mu_{t,j-1}$ is the adjustment for the j -th layer’s overall entropy $\bar{\mathcal{H}}_{t,j}$. The token reduction ratio $\text{Ratio}_{t,j}$ at the t -th scale and j -th layer is:

$$\text{Ratio}_{t,j} = \begin{cases} \text{Ratio}_t, & j = 1, \\ \text{Ratio}_t - \Delta_{t,j} \text{Ratio}_t, & j \geq 2. \end{cases} \quad (13)$$

Finally, pruning low-entropy tokens according to the token reduction ratio can achieve dual-linkage acceleration.

Residual Cache Reuse. As a dense prediction task, visual generation requires preserving a coherent two-dimensional structure, we employ residual cache reuse to compensate for pruned low-entropy tokens. At scales with $t \geq t^*$, the residual cache is activated and updated. The layer transformation at the j -th layer (Transformer layer j) for the t -th scale can be expressed as:

$$R_{t,j}^{\text{output}} = \text{Layer}_j(R_{t,j}^{\text{input}}), \quad (14)$$

Consequently, we construct a layer-wise residual cache:

$$\text{Cache}_{t,j} = R_{t,j}^{\text{output}} - R_{t,j}^{\text{input}}. \quad (15)$$

When predicting the next scale $t+1$ at j -th layer, for those pruned tokens, we interpolate $\text{Cache}_{t,j}$ into $\text{Cache}_{t \rightarrow t+1,j}$

and reconstruct the layer output by reusing residual cache:

$$R_{t,j}^{\text{output}} = \text{Cache}_{t \rightarrow t+1,j} + R_{t,j}^{\text{input}}. \quad (16)$$

By reusing dynamic residual information rather than static tokens, NOVA preserves fine-grained structural cues while avoiding redundant computation on low-entropy tokens.

4. Experiments

4.1. Experimental Settings

Models and Benchmarks. We employ text-to-image VAR models to evaluate NOVA, including Infinity-2B, Infinity-8B (Han et al., 2025), and HART-0.7B (Tang et al., 2024). All selected backbones are capable of high-resolution image generation at 1024×1024 . We utilize GenEval (Ghosh et al., 2023), DPG-Bench (Hu et al., 2024), and MJHQ-30K (Li et al., 2024a) to evaluate generation quality, which are widely adopted for assessing semantic alignment and the perceptual quality of generated images. Moreover, we use ImageReward (Xu et al., 2023) and HPSv2.1 (Wu et al., 2023b). To quantify efficiency, we report the average inference latency and the corresponding speedup ratio achieved by our method.

Implementation Details. To ensure a fair comparison, we evaluate NOVA against all open-source baselines using their default hyperparameter settings. For NOVA, we set the adaptive acceleration activation parameters to $t_{\text{est}} = 5$ and $\alpha = 0.5$, and set the dual-linkage parameters to $\tau = 0.8$ and $\lambda = 0.1$. These hyperparameters are not heavily tuned for each benchmark; we use nearly identical settings across different backbones. The sensitivity analysis in appendix Table 15 and appendix Figure 9 show stable GenEval performance for $\tau \in [0.6, 0.9]$ and $\lambda \in [0.05, 0.25]$, indicating limited tuning overhead. More implementation details are provided in the appendix. All experiments are conducted on

Table 2. Quantitative evaluation on ImageReward and HPSv2.1, two widely used human preference benchmarks.

Methods	Latency	ImageReward		HPSv2.1				
		Score $\uparrow$	CLIP-Score $\uparrow$	Anime $\uparrow$	Concept-Art $\uparrow$	Paintings $\uparrow$	Photo $\uparrow$	Overall $\uparrow$
Infinity-2B	2.43s	0.921	0.27	31.63	30.26	30.28	29.27	30.36
+FastVAR	0.99s	0.903	0.26	31.12	30.15	29.92	28.86	29.95
+SkipVAR	1.09s	0.908	0.27	30.11	30.18	30.13	29.07	30.22
+SparseVAR	1.46s	0.895	0.26	31.22	29.61	29.10	28.21	29.53
+NOVA	0.84s	0.911	0.27	31.40	30.06	30.21	29.11	30.20
Infinity-8B	1.51s	1.030	0.28	32.40	31.17	30.78	29.35	30.92
+FastVAR	0.79s	1.028	0.28	31.80	30.42	29.89	29.87	30.24
+SkipVAR	0.82s	1.032	0.27	32.01	30.58	30.46	29.09	30.44
+NOVA	0.75s	1.035	0.28	32.03	30.71	30.25	29.15	30.54

a single NVIDIA RTX 3090 GPU, except for Infinity-8B, which is evaluated on an NVIDIA H200 GPU due to its larger model size.

4.2. Main Results

Quantitative Performance Comparison. As shown in Table 1 and Table 3, we assess both generation quality and inference efficiency for 1024×1024 image generation on GenEval, DPG-Bench, and MJHQ-30K. Compared with other comparable models, NOVA consistently achieves a superior balance of efficiency and performance. Specifically, for Infinity-2B, NOVA delivers $2.89\times$ speedup while preserving approximately original GenEval Overall score 0.72. Notably, NOVA attains the best performance on Two Object sub-metric, outperforming both the original Infinity-2B and other acceleration methods. Moreover, on DPG-Bench and MJHQ-30K, results also confirm that NOVA preserves high-fidelity visual quality under acceleration. Furthermore, on larger-size and other VAR models such as Infinity-8B and HART, NOVA consistently delivers substantial acceleration, demonstrating its broad applicability and generalizability across diverse model architectures and sizes.

 Table 3. Performance on GenEval $\uparrow$ benchmark.

Methods	Landscape		People		Food	
	FID $\downarrow$	CLIP $\uparrow$	FID $\downarrow$	CLIP $\uparrow$	FID $\downarrow$	CLIP $\uparrow$
HART	25.43	26.82	30.61	28.47	30.37	28.03
+FastVAR	22.52	26.51	28.19	28.34	30.97	28.25
+NOVA	22.81	26.56	28.09	28.33	30.74	28.40
Infinity-2B	29.29	26.18	28.15	27.97	30.73	26.70
+FastVAR	28.08	26.58	27.99	28.31	29.95	27.08
+NOVA	29.29	26.14	27.88	29.29	29.91	27.03

Quantitative Performance on Human Preference. We further evaluate the impact of NOVA’s acceleration on perceptual quality using human preference benchmarks with ImageReward and HPSv2.1. In Table 2, despite the aggressive reduction in token usage, NOVA maintains high align-

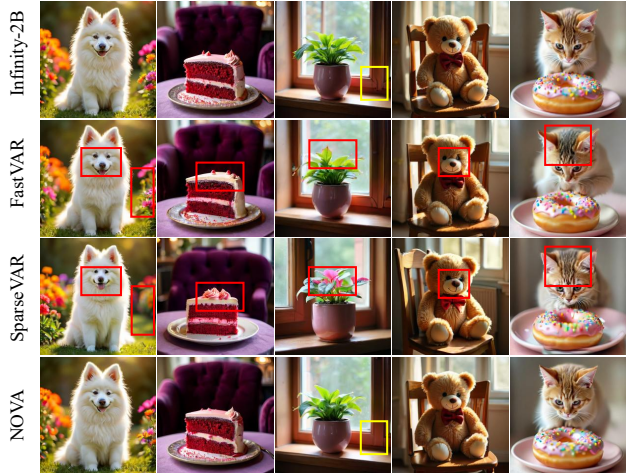


Figure 6. Visual comparison between NOVA and other training-free comparable methods on Infinity-2B. Red and yellow boxes highlight the finer details in visualization.

ment with human aesthetic preferences. Surprisingly, on ImageReward, NOVA obtains a score of 1.035 on Infinity-8B, outperforming the initial Infinity-8B (1.030). This breaks the conventional wisdom that accelerating through token pruning necessarily leads to performance loss. On HPSv2.1, NOVA remains comparable to the unpruned baseline, confirming that the generated images are free of perceptible artifacts and maintain high artistic quality.

Visual Comparison. In Figure 7 and Figure 6, we provide visual comparisons to illustrate the performance of NOVA across different backbones and against comparable token reduction acceleration methods. These results indicate that under the adaptive acceleration activation and dual-linkage acceleration, NOVA achieves near-lossless visual quality mitigates semantic loss, structural distortion, and detail collapse. Notably, in the yellow-box region, NOVA generates clearer and more logically consistent structural details than the original Infinity-2B, aligning better with human preference. More comparisons are provided in the appendix.



Figure 7. Visual comparisons between NOVA and backbones, including Infinity-2B, Infinity-8B, and HART-0.7B.

Table 4. Ablation of token selection metrics on Infinity-2B.

Methods	GenEval $\uparrow$			
	Two Obj.	Position	Color Attri.	Overall
Infinity-2B	0.85	0.45	0.54	0.73
+NOVA w/ Attn	0.84	0.38	0.52	0.69
+NOVA w/ MSE	0.82	0.42	0.52	0.71
+NOVA w/ Entropy	0.86	0.42	0.54	0.72

4.3. Ablation and Analysis

Effectiveness of Entropy Analysis. To examine the effectiveness of our entropy-based method for token reduction, we conduct an ablation study comparing it with Attention-based (Attn) and MSE-based (MSE) alternatives on Infinity-2B. All variants adopt the same dual-linkage configuration for fair comparison. As shown in Table 4, the MSE-based method leads to a noticeable drop in overall GenEval performance, while the entropy-based variant achieves the best overall score among all compared methods. Similar to SparseVAR (Chen et al., 2025b), we treat MSE metric as a proxy for frequency signal magnitude. These results suggest that frequency signals alone may not reliably reflect their impact on potential information, whereas entropy provides a more informative criterion for identifying tokens that contribute important information in the pruning process.

Table 5. Ablation on the effectiveness of adaptive acceleration activation strategy on Infinity-2B. The ‘*’ denote without adaptive acceleration activation.

Methods	Start Scale	Speedup	GenEval $\uparrow$		
			Two Obj.	Color Attri.	Overall
NOVA*	6-th	3.52 $\times$	0.72	0.45	0.62
NOVA*	7-th	2.85 $\times$	0.78	0.49	0.66
NOVA*	8-th	2.14 $\times$	0.81	0.53	0.68
NOVA*	9-th	1.88 $\times$	0.84	0.54	0.70
NOVA*	10-th	1.63 $\times$	0.83	0.54	0.70
NOVA	—	2.89 $\times$	0.86	0.58	0.72

Effectiveness of Adaptive Acceleration Activation. To demonstrate the effectiveness of the adaptive acceleration activation, we compare NOVA with fixed acceleration scales on Infinity-2B. To ensure a fair comparison, all methods adopt the adaptive dual-linkage ratio function to determine reduction ratio at each scale. As shown in Table 5, set the fixed acceleration scale early (e.g., 6-th scale) may lead to

aggressive speedup but causes noticeable quality degradation, whereas set a delaying acceleration scale (e.g., 10-th scale) preserves quality at the cost of reduced acceleration. In contrast, NOVA adaptively determines the acceleration scale, enabling to achieve a more favorable trade-off between aggressive acceleration and quality preservation.

 Table 6. Ablation on effectiveness of dual-linkage acceleration on Infinity-2B. $\dagger$ denotes the variants’ scale-level reduction ratios are fixed following FastVAR, $\ddagger$ denotes the variants’ scale-level reduction ratios following SparseVAR.

Methods	Scale-Linkage	Layer-Linkage	Speedup	GenEval $\uparrow$
NOVA $\dagger$	$\times$	$\times$	2.55 $\times$	0.69
NOVA $\dagger$	$\times$	$\checkmark$	2.34 $\times$	0.70
NOVA $\ddagger$	$\times$	$\times$	2.15 $\times$	0.71
NOVA $\ddagger$	$\times$	$\checkmark$	2.22 $\times$	0.71
NOVA	$\checkmark$	$\times$	2.85 $\times$	0.71
NOVA	$\checkmark$	$\checkmark$	2.89 $\times$	0.72

Effectiveness of Dual-Linkage Acceleration. We conduct an ablation study covering two no-linkage and three single-linkage variants to examine the effectiveness of dual-linkage acceleration. All variants employ adaptive acceleration activation to ensure a fair comparison. As shown in Table 6, both no-linkage variants suffer from an inherent efficiency-quality trade-off. Further, enabling either scale-linkage or layer-linkage alone consistently improves over the corresponding no-linkage variants, while jointly applying dual-linkage pushes this improvement to the best overall efficiency and performance trade-off.

Table 7. Ablation on the effectiveness of residual cache reuse strategy on Infinity-2B. The ‘*’ denotes static token caching strategy.

Methods	Start Cache Scale	Speedup	GenEval $\uparrow$		
			Two Obj.	Color Attri.	Overall
NOVA*	6-th	2.93 $\times$	0.78	0.49	0.67
NOVA*	7-th	2.77 $\times$	0.79	0.45	0.69
NOVA*	8-th	2.33 $\times$	0.81	0.53	0.71
NOVA	—	2.89 $\times$	0.86	0.58	0.72

Effectiveness of Residual Cache Reuse. To verify the effectiveness of the proposed residual cache reuse strategy, we conduct an ablation study on the Infinity-2B backbone. We compare NOVA’s dynamic residual cache reuse with FastVAR’s static token caching, which caches tokens at a predetermined decoding step. For a fair comparison across

cache scales, t_{est} is adjusted accordingly for both methods under comparable acceleration settings. Table 7 shows that dynamic residual cache reuse consistently yields better generation quality under similar or even higher speedup ratios. Unlike static caching, which reuses tokens at fixed decoding steps regardless of the modeling state, NOVA adaptively reuses residual features via entropy-guided token selection. This dynamic behavior better preserves semantic consistency and fine-grained details, while mitigating error accumulation from aggressive token pruning.

Table 8. Latency breakdown of a representative high-resolution scale during VAR-like model inference on the Infinity-2B backbone at 1024×1024 resolution.

Component	Latency (ms)	Ratio (%)
Self-Attention	128.4	71.2
Feed-Forward (FFN)	42.1	23.4
Entropy Computation	1.7	0.9
Others	8.2	4.5
Total	180.4	100.0

Entropy Computation Overhead. We further analyze the runtime overhead introduced by entropy computation. NOVA computes entropy directly from the predictive token distributions already produced during standard decoding, requiring only element-wise operations and reductions. The additional cost scales as $\mathcal{O}(N_t|V|)$ at scale t , which is linear in the number of tokens, while self-attention scales as $\mathcal{O}(N_t^2d)$. In practice, entropy computation takes only 1.7 ms on a representative high-resolution scale of Infinity-2B at 1024×1024 resolution, accounting for 0.9% of the total latency; the full latency breakdown is provided in Appendix C.1. Therefore, the entropy overhead is small and is effectively amortized by the computation saved through token reduction.

Table 9. Quantitative comparison on class-conditional ImageNet generation using VAR-d30 at 256×256 and VAR-d36 at 512×512 resolution. Latency is measured with batch size 5. FID and IS are reported to evaluate generation quality.

Method	Res.	Latency ↓	FID ↓	IS ↑
VAR-d30	256	0.67s	2.05	306.6
VAR-d36	512	2.89s	2.63	303.2
NOVA(VAR-d30)	256	0.63s	2.23	313.2
NOVA(VAR-d36)	512	1.97s	2.79	299.1

Class-Conditional Generation. To examine the generality of NOVA beyond text-to-image generation, we further evaluate it on class-conditional ImageNet generation using VAR-d30 at 256×256 resolution and VAR-d36 at 512×512 resolution, with the batch size fixed to 5. As shown in Table 9, NOVA reduces the latency of VAR-d36 from 2.89s to 1.97s with only a modest change in FID and IS. The acceleration on VAR-d30 is less pronounced, reducing latency from

0.67s to 0.63s, likely because the lower-resolution model has less computational redundancy to exploit. These results suggest that NOVA remains effective in class-conditional generation, especially for higher-resolution settings with larger computational cost.

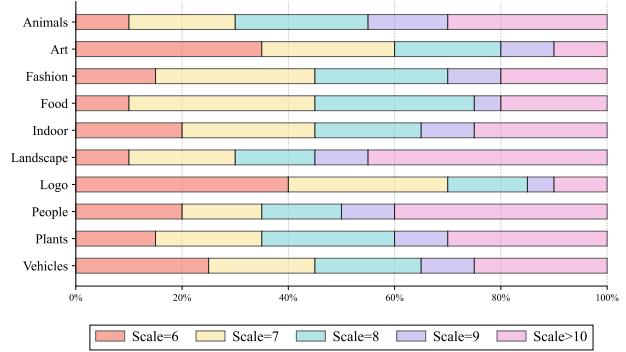


Figure 8. Distribution of acceleration activation scale ratios of NOVA across different categories of the MJHQ-30K benchmark.

Statistical Analysis of Acceleration Activation Scale. We statistically analyze the acceleration activation scale on MJHQ-30K by examining its distribution across image categories. In Figure 8, diverse image categories exhibit distinct activation distributions, indicating that pruning on-set depends on content: fine-grained textures necessitate later pruning, whereas simple global structures allow earlier activation. The results suggest that NOVA can effectively capture context-specific generation characteristics and adaptively identify appropriate acceleration scales, rather than relying on fixed or heuristic reduction schedules. This instance-level variation further supports the need for adaptive acceleration rather than a fixed pruning schedule in VAR inference.

5. Conclusion

This paper introduces NOVA, a training-free acceleration framework for VAR models. By utilizing dual-linkage entropy analysis, NOVA adaptively online identifies the acceleration activation scale through entropy growth inflection point detection. NOVA also dynamically computes distinct token reduction ratios across both scales and layers, pruning low-entropy tokens and reusing residual cache to accelerate inference and maintain quality. Extensive experiments validate NOVA as a simple yet effective acceleration method.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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Appendix

A. More Experiments Details

In this section, we provide additional experimental details to improve the reproducibility of our experiments. Specifically, we categorize the experimental settings into two parts: hyperparameter configurations and ablation study protocols.

A.1. Hyperparameter Settings

We report all hyperparameters used in NOVA to facilitate reproducibility. As shown in Table 10, these include the entropy-based adaptive acceleration activation parameters (t_{est} , and α), as well as the dual-linkage token reduction parameters (τ and λ). Unless otherwise stated, these hyperparameters are fixed across all experiments and applied consistently to all evaluated backbones. For baseline methods, we strictly follow the default hyperparameter settings released in their official implementations and use the same set of random seeds as in our experiments to ensure a fair and reproducible comparison.

Applying NOVA to new domains or backbones does not require extensive hyperparameter search. The activation parameters t_{est} and α control when acceleration starts, while τ and λ adjust the token reduction ratios across scales and layers. For 1024px generation with 13 scales, we recommend $t_{\text{est}} \in [4, 6]$, $\alpha \in [0.4, 0.6]$, $\tau = 0.8$, and $\lambda = 0.1$ as default settings. A more aggressive setting can be obtained by decreasing α and slightly increasing λ , whereas a more conservative setting can be obtained by increasing α and decreasing λ . The cross-backbone results in Table 10 and the sensitivity analyses indicate that NOVA is robust to moderate variations of these parameters.

Table 10. Hyperparameter settings of NOVA across different VAR-like backbone models. All hyperparameters are kept identical unless otherwise specified.

Backbone	t_{est}	α	τ	λ
Infinity-2B	5	0.5	0.8	0.1
Infinity-8B	5	0.5	0.8	0.1
HART-0.7B	6	0.4	0.7	0.1
VAR-d36	4	0.4	0.8	0.1

A.2. Ablation Protocol

All ablation studies are conducted under a unified and controlled experimental protocol. Unless otherwise specified, we perform ablations on the Infinity-2B backbone and modify only one component or hyperparameter at a time, while keeping all other settings unchanged.

Effectiveness of Entropy-based Method. To evaluate the effectiveness of the token selection criterion, we replace the entropy-based method with MSE-based and attention-based variants while keeping all other components unchanged. For the MSE-based variant, token importance is measured by the magnitude of feature variation at each Transformer layer. For the n -th layer at scale t , the importance score of token i is computed as:

$$s_{t,i}^n = \|X_{t,\text{output},i}^n - X_{t,\text{input},i}^n\|_2^2, \quad (17)$$

where $X_{t,\text{input},i}^n$ and $X_{t,\text{output},i}^n$ denote the input and output representations of token i at layer n .

In addition, we construct an attention-based variant, where token importance is estimated from attention responses. For the attention-based variant, at each layer n and scale t , we compute the average attention weight received by token i across all attention heads as:

$$s_{t,i}^n = \frac{1}{H} \sum_{h=1}^H \sum_j A_{t,j \rightarrow i}^{n,h}, \quad (18)$$

where $A_{t,j \rightarrow i}^{n,h}$ denotes the attention weight from token j to token i in head h , and H is the number of attention heads.

Effectiveness of Dual-Linkage adaptive Token Reduction Method. For this ablation, we evaluate two no-linkage variants and three single-linkage variants under the same experimental setting. All variants use Adaptive Acceleration Activation to determine the acceleration start scale. For the no-linkage variants, token reduction ratios are fixed once acceleration is activated. The variant marked with $\dagger$ follows the scale-level pruning ratios used in FastVAR, where the reduction ratios are set to:

$$\{20:0.10, 24:0.15, 32:0.40, 40:0.50, 48:0.90, 60:0.95\}.$$

These ratios are applied uniformly across all layers.

The variant marked with $\ddagger$ follows the pruning strategy of SparseVAR, where the token reduction ratio at each scale is determined by the token importance score defined in Eq. 17, without scale or layer linkage. For the single-linkage variants, we enable only one linkage dimension at a time. In the scale-linkage variant, the token reduction ratio is adaptively adjusted across scales, while remaining identical across layers. In the layer-linkage variant, the reduction ratio is adaptively determined across layers, while the scale-level ratio is fixed. The dual-linkage variant enables both scale-linkage and layer-linkage.

B. More Experiments Results

Table 11. Quantitative comparisons of perceptual quality on the GenEval and DPG-Bench benchmarks.

Methods	Speed $\uparrow$	Latency $\downarrow$	Param $\downarrow$	GenEval $\uparrow$				DPG-Bench $\uparrow$		
				Two Obj.	Position	Color Attri.	Overall	Global	Relation	Overall
SDXL	-	4.3s	2.6B	0.74	0.15	0.23	0.55	83.27	86.76	74.65
LlamaGen	-	37.7s	0.8B	0.34	0.07	0.04	0.32	-	-	65.16
Show-o	-	50.3s	1.3B	0.80	0.31	0.50	0.68	-	-	67.48
PixArt-Sigma	-	2.7s	0.6B	0.62	0.14	0.27	0.55	86.89	86.59	80.54
DALL-E 3	-	-	-	-	-	-	0.67	90.97	90.58	83.50
Emu3	-	-	8.5B	0.81	0.49	0.45	0.66	-	-	81.60
HART	$1.00 \times$	1.12s	0.7B	0.62	0.13	0.18	0.51	87.10	92.95	80.89
FastVAR	$1.30 \times$	0.86s	0.7B	0.57	0.16	0.19	0.50	86.21	92.48	80.71
SparseVAR	$1.38 \times$	0.81s	0.7B	0.60	0.12	0.21	0.50	86.63	92.54	80.70
NOVA	$1.62 \times$	0.69s	0.7B	0.60	0.13	0.17	0.51	86.88	92.53	80.84
Infinity-2B	$1.00 \times$	2.43s	2.0B	0.85	0.45	0.54	0.73	85.10	92.37	83.12
FastVAR	$2.45 \times$	0.99s	2.0B	0.81	0.37	0.52	0.70	84.41	92.76	82.86
SparseVAR	$1.66 \times$	1.46s	2.0B	0.84	0.42	0.54	0.72	84.11	92.70	82.56
SkipVAR	$2.23 \times$	1.09s	2.0B	0.84	0.39	0.60	0.72	84.19	93.07	82.94
NOVA	$2.89 \times$	0.84s	2.0B	0.86	0.42	0.58	0.72	84.36	92.68	82.93
Infinity-8B	$1.00 \times$	1.51s	8.0B	0.94	0.58	0.68	0.80	85.10	94.50	86.60
FastVAR	$1.26 \times$	0.79s	8.0B	0.92	0.55	0.68	0.78	84.80	92.30	86.46
SkipVAR	$1.22 \times$	0.82s	8.0B	0.91	0.57	0.69	0.77	83.91	90.52	86.40
NOVA	$1.33 \times$	0.75s	8.0B	0.92	0.58	0.69	0.79	84.27	93.61	86.53

Comparison on GenEval and DPG-Bench. To provide a more comprehensive evaluation, we further compare NOVA with a broader range of representative methods, as shown in Table 11. In addition to VAR-like models, this comparison includes autoregressive models based on next-token prediction and state-of-the-art diffusion-based models. The results demonstrate that NOVA achieves strong and competitive performance across different model families. Notably, beyond consistently outperforming VAR-like baselines, NOVA also remains highly competitive against other paradigms, achieving comparable performance to state-of-the-art diffusion models.

Results with PSNR, SSIM, and LPIPS. To assess potential semantic degradation introduced by acceleration, we further conduct an additional evaluation on the MJHQ-30K benchmark using the Infinity-2B backbone. Specifically, we compare generation results with and without applying NOVA under the same inference settings. As shown in Table 12, NOVA maintains comparable semantic quality while substantially improving inference efficiency. Consistent with the main results,

Table 12. Quantitative comparison on different MJHQ subsets using the Infinity-2B backbone. Metrics are reported as PSNR $\uparrow$, SSIM $\uparrow$, and LPIPS $\downarrow$.

Methods	Food			Landscape			People		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
FastVAR	14.97	0.59	0.40	17.67	0.54	0.40	16.38	0.58	0.41
NOVA	14.98	0.60	0.40	17.74	0.55	0.39	16.39	0.60	0.39

NOVA achieves competitive performance across all metrics while substantially improving inference efficiency, indicating that the proposed acceleration does not compromise low-level visual fidelity.

Table 13. Comparison between NOVA and ToMe on the HART-0.7B backbone.

Method	Speedup $\uparrow$	GenEval $\uparrow$
HART-0.7B	1.00 $\times$	0.51
ToMe	1.19 $\times$	0.48
ToMe	1.36 $\times$	0.46
NOVA	1.62 $\times$	0.51

Comparison with ToMe. Token Merging (ToMe (Bolya et al., 2022)), which improves inference efficiency by merging multiple tokens during Transformer computation, is a popular model acceleration method. We further compare NOVA with ToMe under the same experimental setting. Specifically, we evaluate both methods on the HART-0.7B backbone and report their GenEval scores for comparison, as shown in Table 13. The results indicate that in VAR-like autoregressive image generation, token merging methods with limited compression, such as ToMe, suffer from more pronounced information loss. For example, ToMe achieves a 1.36 $\times$ speedup but underperforms NOVA by approximately 10% on GenEval, despite NOVA attaining a higher 1.62 $\times$ acceleration ratio. These results suggest that combining token merging and token dropping strategies may serve as a promising direction for future research in efficient VAR-like autoregressive image generation.

C. More Ablation and Analysis

C.1. Entropy Computation Overhead Analysis.

To further justify the practical efficiency of NOVA, we analyze the computational overhead introduced by entropy calculation during inference. In VAR-like models, the scale-level entropy is computed directly from the token-level predictive distributions, which are already produced by the softmax operation in standard decoding. Consequently, entropy calculation only involves lightweight element-wise operations and reductions, without introducing additional attention layers or matrix multiplications.

From a complexity perspective, the computational cost of entropy calculation at scale t is

$$\mathcal{C}_{\text{entropy}}(t) = \mathcal{O}(N_t \cdot |V|), \quad (19)$$

where N_t denotes the number of tokens at scale t and $|V|$ is the vocabulary size. In contrast, the dominant computational cost in VAR inference arises from self-attention, whose complexity scales quadratically with the token count:

$$\mathcal{C}_{\text{attn}}(t) = \mathcal{O}(N_t^2 \cdot d), \quad (20)$$

where d is the hidden dimension. Compared to the quadratic complexity of self-attention, the linear cost of entropy computation is asymptotically negligible.

As a result, the overhead introduced by entropy calculation does not undermine the overall inference efficiency. Instead, it is effectively amortized by the substantial reduction in attention and feed-forward computation enabled by entropy-guided token pruning in NOVA, while preserving a clean and modular inference pipeline without introducing additional architectural complexity.

We further measure the wall-clock latency of entropy computation during inference. As shown in Table 14, entropy-related operations account for less than a small fraction of the total inference time, while enabling substantial token reduction that significantly decreases attention and feed-forward computation. Therefore, the computational cost introduced by entropy estimation is effectively amortized by the resulting reduction in token-level computation, and does not undermine the overall acceleration benefits of NOVA.

Table 14. Latency breakdown of a representative high-resolution scale during VAR-like model inference on the Infinity-2B backbone at 1024×1024 resolution.

Component	Latency (ms)	Ratio (%)
Self-Attention	128.4	71.2
Feed-Forward (FFN)	42.1	23.4
Entropy Computation	1.7	0.9
Others	8.2	4.5
Total	180.4	100.0

C.2. Ablation on Hyperparameters of Dual-Linkage Acceleration.

We conduct an ablation study to analyze the sensitivity of key hyperparameters in the proposed dual-linkage acceleration strategy.

Table 15. Ablation study investigating the impact of the scale-linkage smoothing parameter τ on the Infinity-2B backbone. τ controls the transition steepness of the token reduction ratio. The default setting is $\tau = 0.8$.

τ	Speedup	GenEval $\uparrow$			
		Two Obj.	Position	Color Attri.	Overall
0.6	$3.05 \times$	0.83	0.40	0.55	0.70
0.7	$2.96 \times$	0.85	0.41	0.57	0.71
0.8 (Default)	$2.89 \times$	0.86	0.42	0.58	0.72
0.9	$2.72 \times$	0.86	0.43	0.58	0.72

Impact of Hyperparameter τ in Scale-Linkage Acceleration. In the Dual-Linkage Acceleration mechanism of NOVA, the parameter τ serves as the temperature scaling factor in Eq 11, controlling the smoothness of the token reduction ratio transition across scales. A smaller τ induces a steeper transition, leading to a rapid increase in pruning ratios, whereas a larger τ results in a more gradual curve. To investigate the sensitivity of this hyperparameter and its impact on the efficiency-quality trade-off, we conducted an ablation study on the Infinity-2B backbone by varying $\tau \in \{0.6, 0.7, 0.8, 0.9\}$ while keeping other settings fixed. As shown in Table 15, NOVA demonstrates strong robustness to variations in τ , with the Overall GenEval score fluctuating stably between 0.70 and 0.72. Overall, while τ influences the speed-quality trade-off, its performance is relatively insensitive to moderate variations, and thus does not introduce significant hyperparameter tuning overhead.

Sensitivity of Refinement Factor λ . We investigate the impact of the scaling factor λ in the scale-linkage ratio as proposed in Eq. 11, which refines the base pruning ratio according to the entropy growth rate g_t . We vary λ from 0.05 to 0.35 with a step of 0.05. As illustrated in Figure 9, the GenEval score remains relatively stable within the range of $[0.05, 0.25]$, consistently maintaining an Overall score of approximately 0.72. This demonstrates that NOVA is robust to the intensity of entropy-based refinement within a broad margin. However, a significant performance degradation is observed when $\lambda > 0.25$, indicating that excessive refinement disrupts the optimal balance of token allocation between scales. Consequently, we select $\lambda = 0.1$ as the default value to ensure both stability and high generation quality.

D. More Visualization Results

In this section, we present additional visualizations that complement the observations described in the main text.

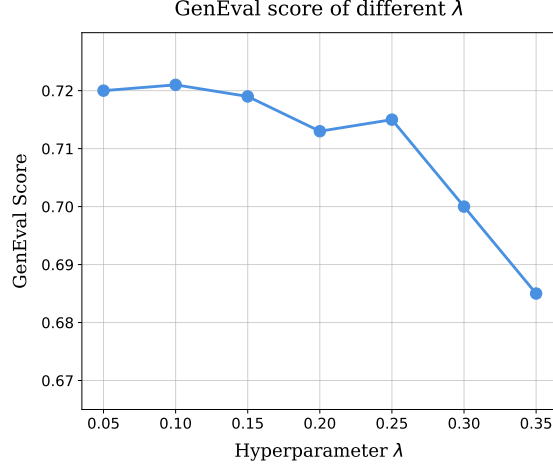


Figure 9. Ablation study of the refinement factor λ on the Infinity-2B backbone. The GenEval score remains stable within a wide range and starts to decline when λ exceeds 0.25.

More Entropy Heatmap Visualization. As shown in Figure 10, we provide additional visualizations of entropy heatmaps of token maps across different scales and Transformer layers. These visualizations reveal clear cross-scale and cross-layer heterogeneity in how the model allocates uncertainty and computational focus during the autoregressive generation process.

Importantly, these cross-scale and cross-layer entropy variations are instance-dependent, as illustrated by the diverse entropy evolution patterns observed across different images. Such heterogeneity indicates that treating all scales or layers uniformly is suboptimal, and motivates the design of NOVA to adaptively allocate computation through entropy-guided, dual-linkage token reduction across both scales and layers.

More Comparison Visualization. We provide additional qualitative comparisons of generated images before and after acceleration to further illustrate the effectiveness of NOVA. As shown in Figure 11 and Figure 12, we present visual comparisons on both Infinity-2B and Infinity-8B backbones. Across different model scales, NOVA consistently achieves effective inference acceleration while preserving high visual fidelity.

Specifically, the accelerated results maintain coherent semantic structures and fine-grained details that are comparable to the unaccelerated baselines. Notably, NOVA effectively mitigates common degradation issues observed in aggressive acceleration settings, such as semantic misalignment and detail collapse. These qualitative results demonstrate that NOVA generalizes well across different VAR backbones, achieving a favorable trade-off between efficiency and generation quality.

High Quality of NOVA Visualization We provide a series of high-quality visualizations to further demonstrate the visual fidelity preserved by NOVA under acceleration. Figure 13 presents generated images with fixed resolution ratios, illustrating that NOVA maintains coherent semantic structures and fine-grained details at a consistent resolution. In contrast, Figure 14 shows visualizations under different resolution ratios, highlighting NOVA’s robustness to resolution variation while preserving visual quality.

E. Limitations

Heuristic Design. As a training-free acceleration framework, NOVA inevitably involves heuristic design choices for acceleration activation and ratio scheduling, such as the entropy-growth threshold and smoothing strategy. Although our ablations show that NOVA is robust to moderate hyperparameter changes, the resulting schedule is not guaranteed to be globally optimal for every prompt or backbone.

Limited Acceleration Scope. NOVA preserves the early generation scales to avoid disrupting global structure formation, and therefore cannot fully exploit potential redundancy at the earliest stages. This limitation is shared by methods that avoid immediate token reduction from the first scale, although NOVA expands the acceleration scope by adaptively activating reduction earlier than fixed late-stage schedules when appropriate.

Peak Memory Reduction. NOVA mainly reduces token-level computation in attention and feed-forward operations, but does not directly optimize the KV cache, which is a major contributor to peak GPU memory usage in VAR-like models. As a result, NOVA can partially alleviate memory consumption through token reduction, but peak memory reduction is not its primary optimization target.



Figure 10. More entropy heatmap visualizations across different scales and Transformer layers on the Infinity-2B backbone. The entropy heatmaps illustrate the spatial distribution and evolution of predictive uncertainty, revealing distinct cross-scale and cross-layer patterns associated with global structure formation and fine-grained texture refinement.

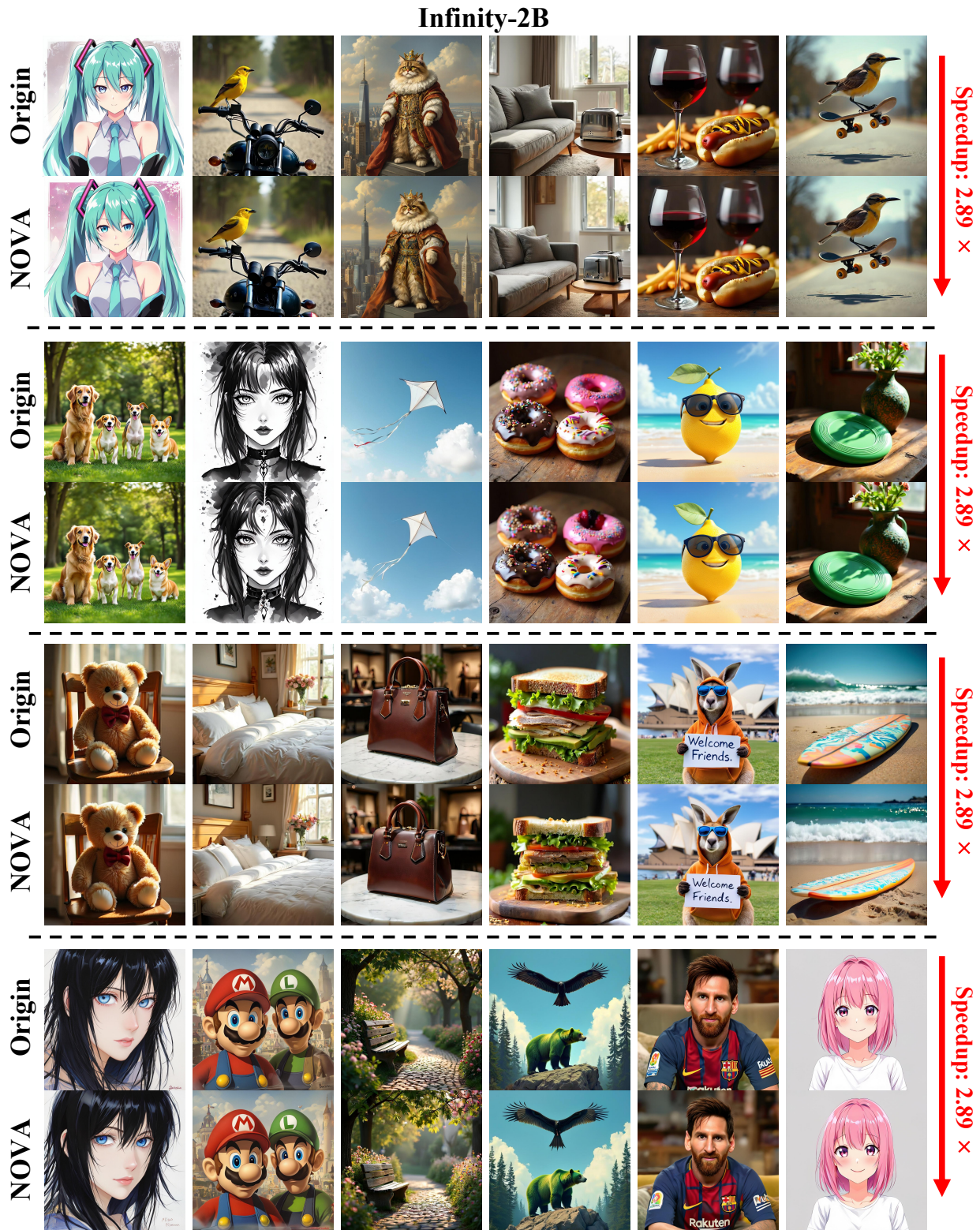


Figure 11. Additional qualitative comparisons between unaccelerated and NOVA-accelerated image generation on the Infinity-2B.

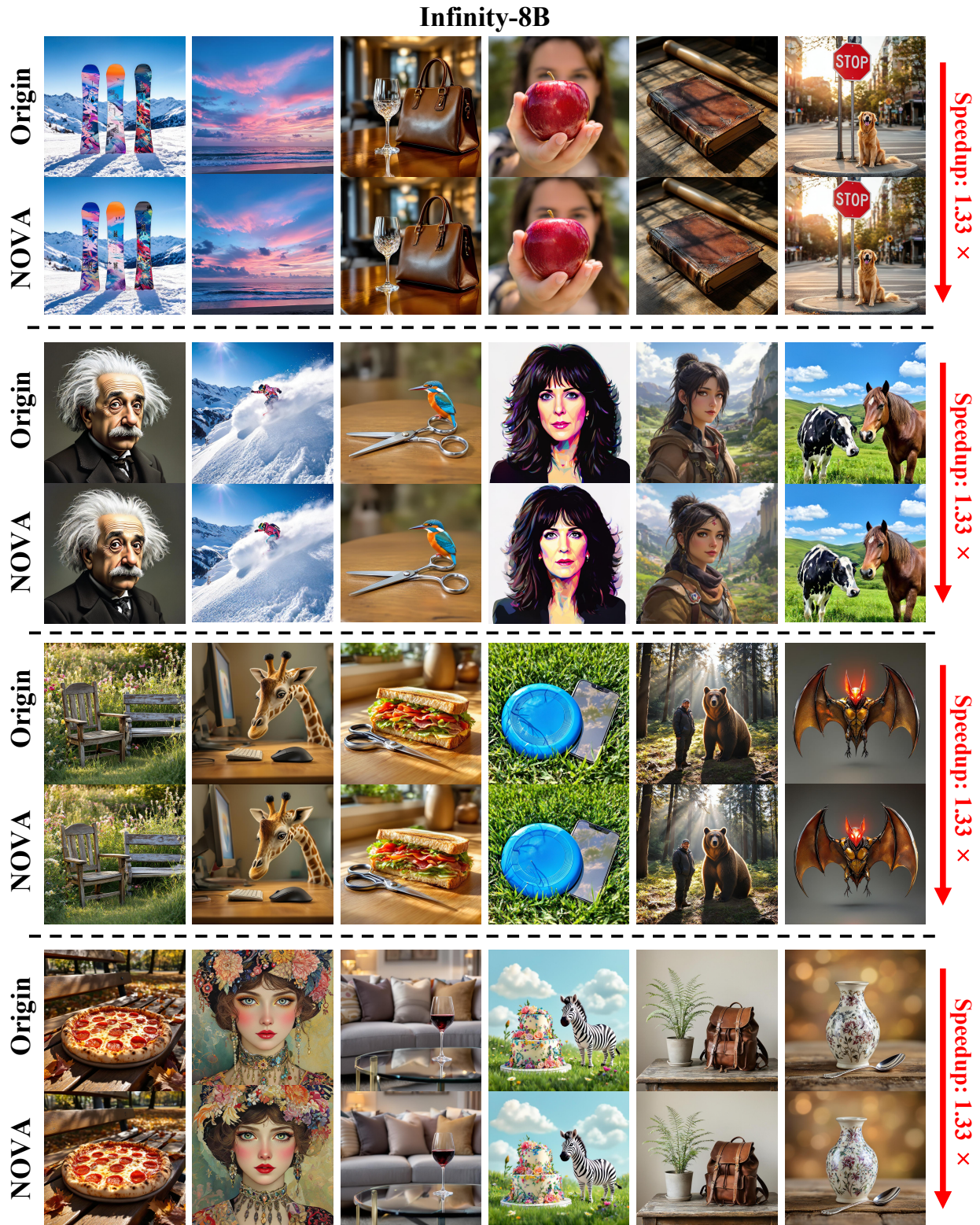


Figure 12. Additional qualitative comparisons between unaccelerated and NOVA-accelerated image generation on the Infinity-8B.

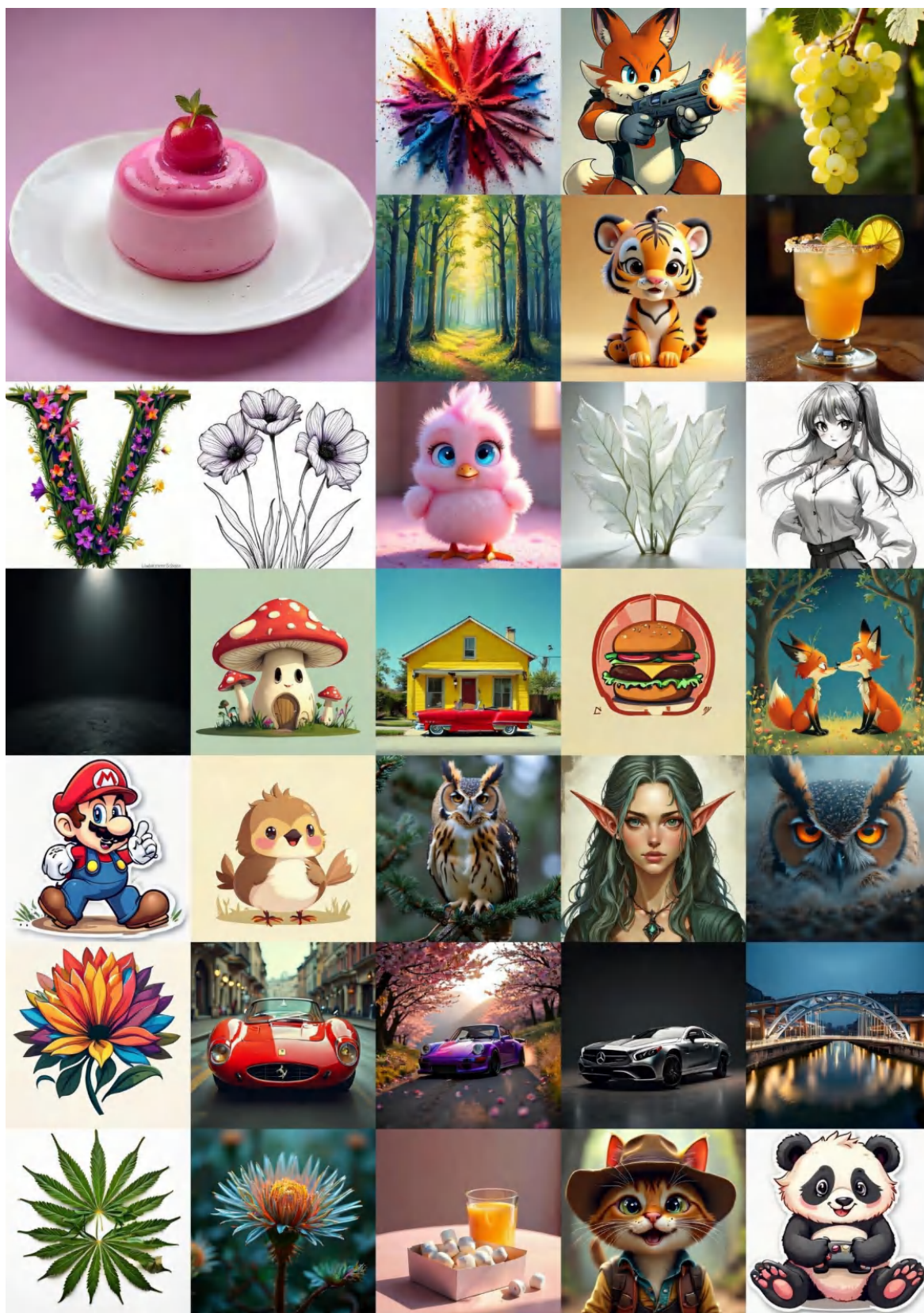


Figure 13. High-quality image generation with fixed resolution ratios using NOVA.

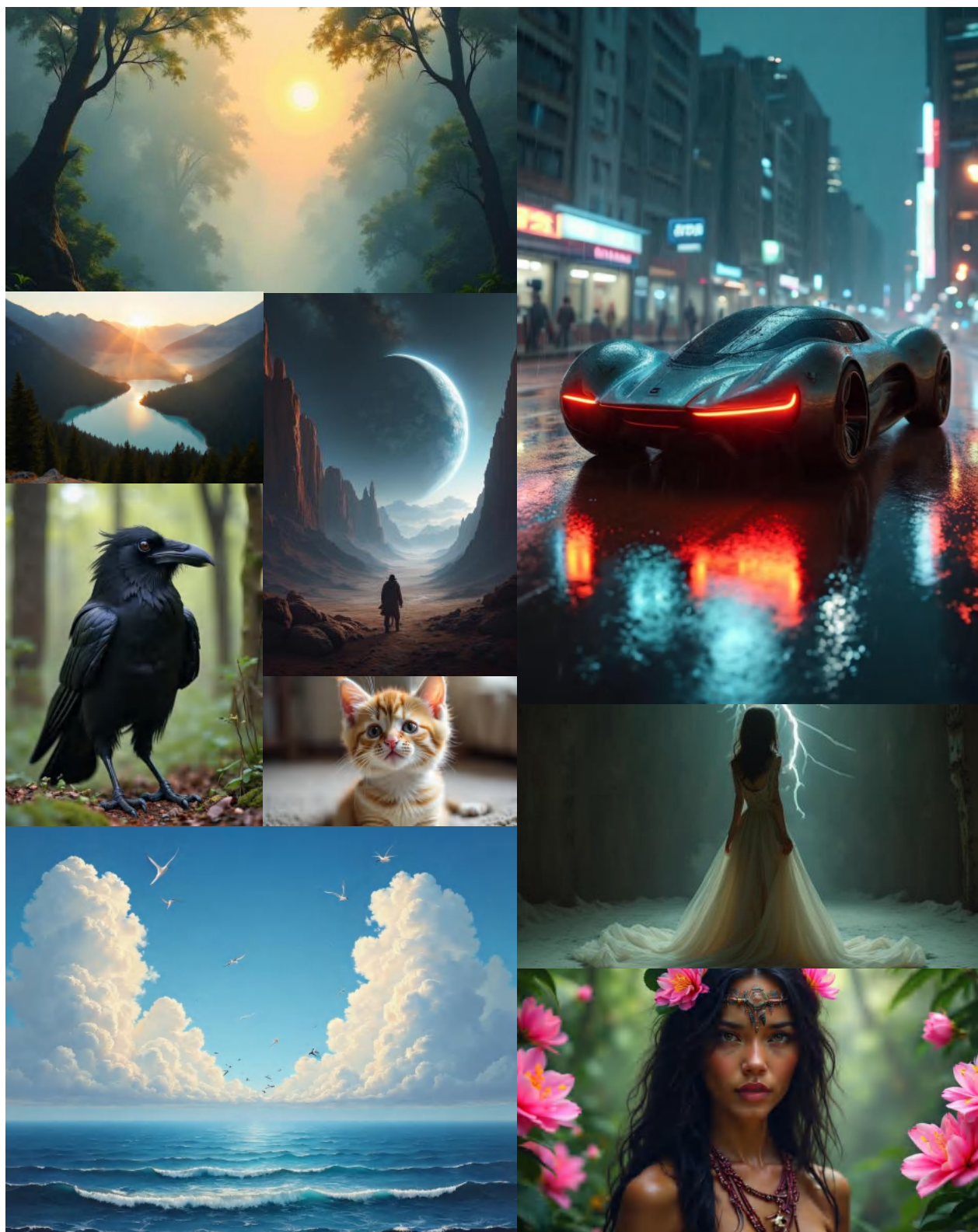


Figure 14. High-quality image generation under different resolution ratios using NOVA.